

5. Conclusion

To meet the demands for high-resolution XR displays, we have successfully developed a micro OLED display with a pixel density exceeding 8000 ppi. Despite the extremely small pixel size for 8000 ppi, stable characteristics of crystal IO FETs enable display operation without significant issues. The combination with MML technology also eliminates color mixing problems.

Meanwhile, there is also a demand for displays of 1 inch or larger sizes for VR applications. Although our prototype in this work achieved a pixel density exceeding 8000 ppi, the display size is 0.5 inches. Scaling up to 1 inch or larger sizes while maintaining this pixel density would result in over $4K \times 4K$ pixels, potentially leading to challenges in processing speed and power consumption. To tackle these challenges, we plan to explore adopting a display driver that can achieve our proposed foveated mode [24] and developing a low-power display by leveraging the low off-leakage current characteristics of crystal IO in the future.

6. References

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